

MERIDIAN & ASSOCIATES LLP

Advisory | Assurance | Tax | Consulting

Case Study: Supply Chain Transformation

Global Industrial Manufacturer

CLIENT CONFIDENTIAL - BLINDED

Version 1.1 | Effective Date: October 2025

Meridian & Associates LLP | New York | Chicago | San Francisco | London | Singapore

This document is the proprietary and confidential property of Meridian & Associates LLP. It is intended solely for the use of the designated recipient(s). Any unauthorized reproduction, distribution, or disclosure of this material is strictly prohibited. The information contained herein is provided for informational purposes only and does not constitute a binding offer or agreement.

1. Engagement Overview

Client: Fortune 200 diversified industrial manufacturer (name withheld)

Industry: Manufacturing - Diversified Industrials

Engagement Type: Supply Chain Transformation

Duration: 16 months (March 2024 - June 2025)

Team Size: 65 professionals at peak staffing

Total Fees: \$18.2 million

Lead Partner: James O'Sullivan, CPA, Six Sigma Black Belt

Meridian Offices: Chicago (lead), San Francisco, Hyderabad (GDC)

2. Client Situation & Challenge

The client, a Fortune 200 diversified industrial manufacturer with \$18.6 billion in annual revenue, operated a global supply chain spanning 14 countries, 22 manufacturing plants, 8 distribution centers, and a network of 3,400 direct material suppliers. Despite significant investments in manufacturing technology and capacity, the company's supply chain performance had deteriorated significantly over the preceding three years, driven by:

- **Excess Inventory:** Finished goods and work-in-progress inventory had ballooned to \$380M, representing 74 days of supply versus an industry benchmark of 45-50 days. Carrying costs alone exceeded \$45M annually. The excess was concentrated in slow-moving SKUs and safety stock buffers inflated during the COVID-era supply disruptions but never rationalized.
- **Poor Demand Visibility:** The company used a fragmented collection of spreadsheet-based forecasting processes across its 5 business units, with no integrated demand planning platform. Forecast accuracy at the SKU-location level averaged 52%, well below the 70-75% industry benchmark, resulting in simultaneous overstock and stockout conditions across the distribution network.
- **Delivery Performance Deterioration:** On-Time-In-Full (OTIF) delivery had declined from 91% to 82% over three years, directly impacting customer satisfaction and contributing to \$28M in customer penalties and lost contract renewals. The root cause analysis revealed fragmented order promising logic, disconnected transportation planning, and lack of real-time inventory visibility across facilities.
- **Logistics Cost Escalation:** Total logistics costs (transportation, warehousing, and distribution) had increased by 34% over three years to \$412M annually, driven by expedited shipping to compensate for poor planning, suboptimal carrier utilization, and a distribution network that had not been redesigned since 2015.
- **Supply Chain Sustainability Gap:** The company had committed to Science Based Targets (SBTi) for Scope 3 emissions reduction but lacked visibility into supplier emissions data. Only 12 of their top 200 suppliers had provided emissions data, creating a significant ESG reporting and compliance risk.

3. Meridian's Solution

Meridian designed and delivered a holistic supply chain transformation program comprising four integrated workstreams:

3.1 SAP IBP Implementation

Meridian implemented SAP Integrated Business Planning (IBP) as the unified demand and supply planning platform across all 5 business units. The implementation covered Demand Planning, Supply Planning, Inventory Optimization, and Sales & Operations Planning (S&OP) modules. The team configured statistical forecasting models calibrated to the client's product hierarchy (15,000 active SKUs), integrated point-of-sale data from 8 key retail customers, and designed a monthly S&OP process connecting commercial demand signals to manufacturing capacity planning.

A critical innovation was the integration of external demand signals, including commodity price indices, weather patterns, and macroeconomic indicators, into the IBP demand sensing engine. This AI-augmented approach improved short-term (4-week) forecast accuracy from 52% to 78%, significantly reducing the need for safety stock buffers.

3.2 Supply Chain Control Tower

Meridian designed and deployed a digital Supply Chain Control Tower providing real-time end-to-end visibility from supplier shipments through manufacturing to customer delivery. Built on a combination of SAP Digital Supply Chain and custom analytics layers, the control tower integrated data from 14 source systems including ERP, TMS, WMS, and supplier portals. Key capabilities included:

- Real-time inventory visibility across all 22 plants and 8 distribution centers (updated every 15 minutes)
- Automated exception management with ML-based anomaly detection (identifying potential disruptions 5-7 days in advance)
- Dynamic order promising engine incorporating real-time ATP (Available-to-Promise) across the network
- Transportation visibility with GPS tracking integration for 85% of shipment volume
- Executive dashboards with drill-down capability from enterprise KPIs to individual shipment level

3.3 Supplier Collaboration Portal

To address supplier visibility gaps, Meridian built a cloud-based Supplier Collaboration Portal using SAP Ariba and custom extensions. The portal enabled real-time purchase order collaboration, advance ship notice (ASN) exchange, quality certificate submission, and capacity commitment sharing. Within 6 months of launch, 280 of the top 300 suppliers (93%) were actively using the portal, reducing purchase order cycle time from 5.2 days to 1.8 days and improving supplier on-time delivery from 78% to 91%.

3.4 Sustainability & Scope 3 Emissions Visibility

Meridian integrated an ESG data collection module into the Supplier Collaboration Portal, enabling standardized Scope 3 emissions data collection aligned with the GHG Protocol. The team developed supplier-specific emissions factors for the top 200 suppliers (representing 82% of procurement spend) using a combination of primary data collection and industry-average estimation models. This created the foundation for the client's first comprehensive Scope 3 emissions baseline, enabling target-setting and progress tracking against SBTi commitments.

4. Delivery Approach

The 16-month program was delivered in three phases:

- **Phase 1 - Design & Foundation (Months 1-4):** Current state assessment, solution architecture, IBP system design, control tower requirements, supplier portal MVP design
- **Phase 2 - Build & Deploy (Months 5-12):** IBP configuration and testing, control tower development, supplier onboarding (3 waves), network optimization analysis, change management
- **Phase 3 - Optimize & Scale (Months 13-16):** ML model tuning, advanced analytics deployment, remaining supplier onboarding, benefits realization, knowledge transfer

Team composition at peak (65 professionals):

- 1 Partner, 3 Senior Managers, 6 Managers, 18 Senior Consultants, 22 Consultants, 15 Analysts
- 45 onshore (Chicago, client sites), 20 offshore (Hyderabad GDC)
- Embedded at 3 client manufacturing sites and 2 distribution centers

5. Results & Impact

Metric	Baseline	Result	Improvement
Inventory Level	\$380M (74 DOS)	\$274M (48 DOS)	28% reduction
Working Capital Freed	-	\$106M	One-time release
OTIF Delivery	82%	95%	+13 points
Forecast Accuracy (4-wk)	52%	78%	+26 points
Logistics Cost	\$412M/year	\$338M/year	18% reduction
Supplier On-Time Delivery	78%	91%	+13 points
Scope 3 Data Coverage	6% of suppliers	Top 200 (82% spend)	Full visibility

6. Client Testimonial

"Meridian transformed our supply chain from a cost center and constant source of frustration into a genuine competitive advantage. The \$106 million in freed working capital alone more than justified the investment, but the real game-changer has been the visibility -- for the first time, we can see our entire supply chain in real time and make decisions proactively rather than reactively. Our customers have noticed the difference."

-- Chief Supply Chain Officer, Client Organization